

ARTIFICIAL INTELLIGENCE II





CHAPTER-5

Expert System





What is an expert system?

An expert system compared with traditional computer :

Inference engine + Knowledge = Expert system

(Algorithm + Data structures = Program in traditional computer)



Steps in Expert System Development





Knowledge Base

- Stores all relevant information, data, rules, cases, and relationships used by the expert system.

Uses

1. Rules
2. If-then Statements
3. Fuzzy Logic

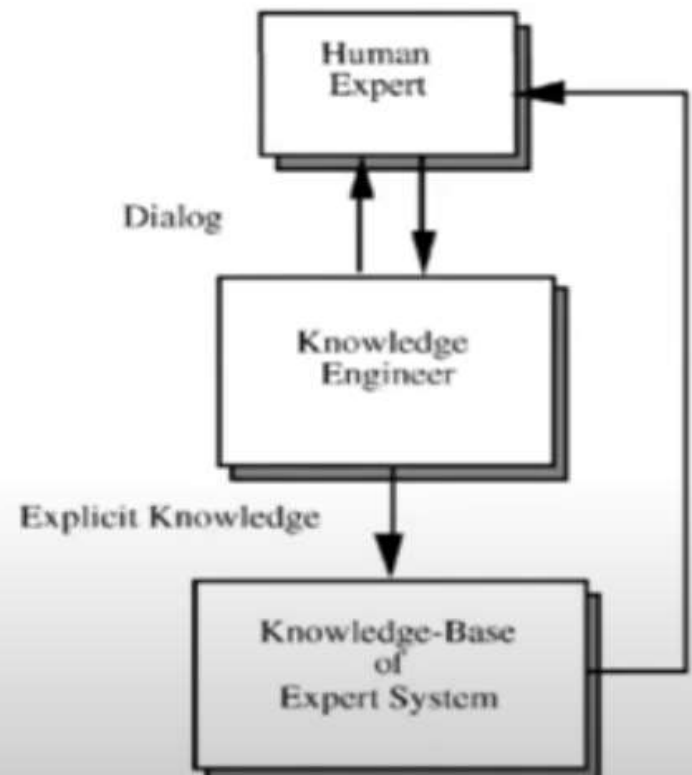
Knowledge acquisition facility

- Provide convenient and efficient means of capturing and storing all the components of the knowledge base.
- Acts as an interface between experts and the knowledge base.



Knowledge Base

- The **knowledge engineer** establishes a dialog with the human expert to elicit knowledge.
- The **knowledge engineer** **codes the knowledge explicitly** in the knowledge base.
- The **expert** **evaluates** the expert system and gives a critique to the knowledge engineer.





Inference Engine

- Seeks information and relationships from the knowledge base and provides answers, predictions, and suggestions the way a human expert would.

Uses

1. Backward Chaining
2. Forward Chaining

Explanation Facility

- Allows a user to understand how the expert system arrived at certain conclusions or results.
- For example: it allows a doctor to find out the logic or rationale of the diagnosis made by a medical expert system.



User Interface

- Specialized user interface software employed for designing, creating, updating, and using expert systems.
- The main purpose of the user interface is to make the development and use of an expert system easier for users and decision makers



Early Expert Systems

- **DENDRAL** – used in chemical mass spectroscopy to identify chemical constituents
- **MYCIN** – medical diagnosis of illness
- **DIPMETER** – geological data analysis for oil
- **PROSPECTOR** – geological data analysis for minerals
- **XCON/R1** – configuring computer systems

× ○ DIGITAL LEARNING CONTENT



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